Exercise 1: Implementing the Singleton Pattern

CODE :

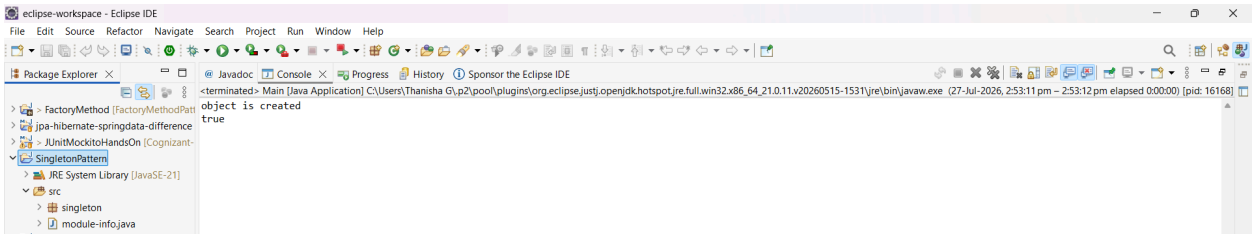
LOGGER.JAVA :

```
package singleton;

public class Logger {
	private static Logger instance;
	private Logger(){
		System.out.println("object is created");
	}
	public static Logger getInstance() {
		if(instance == null) {
			instance = new Logger();
		}
		return instance;
	}

}
```

MAIN.JAVA :

```
package singleton;

public class Main {

	public static void main(String[] args) {

		Logger log1 = Logger.getInstance();
		Logger log2 = Logger.getInstance();
		System.out.println(log1==log2);
		}

}
```

OUTPUT :



Exercise 2: Implementing the Factory Method Pattern

EmailFactory .java :

```
package factory;

public class EmailFactory implements Notification{



@Override
public void send() {

System.out.println(
        "Sending  Email Notification");

}

}
```

EmailNotification.java :

```
package factory;

public class EmailNotification
                implements Notification {

            @Override
            public void send() {

                System.out.println(
                        "Sending Email Notification");

            }
        }
```

SMSFACTORY.JAVA :

```
package factory;

public class SMSFactory
      extends NotificationFactory {

   @Override
   public Notification createNotification() {

      return new SMSNotification();

   }
}
```

SMSNOTIFICATION .JAVA

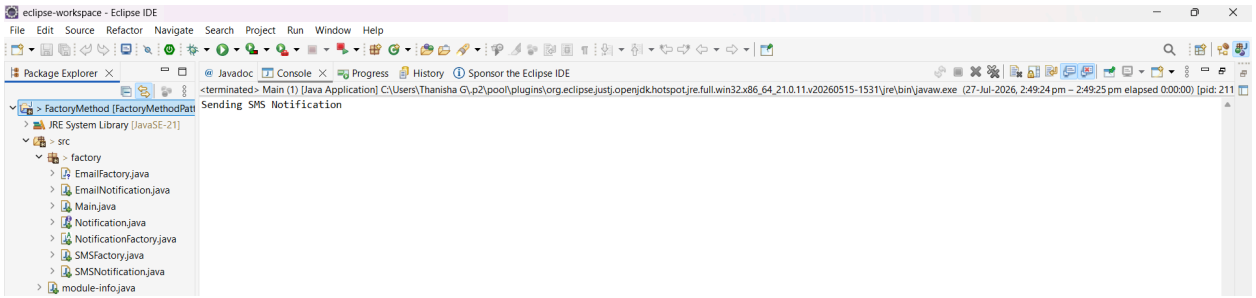
```
package factory;

public class SMSNotification
implements Notification {

@Override
public void send() {

System.out.println(
      "Sending SMS Notification");

}

}
```

MAIN.JAVA

```
package factory;

public class Main {

        public static void main(String[] args) {
                NotificationFactory factory = new SMSFactory();
                Notification notification = factory.createNotification();
                notification.send();

        }
```

}

OUTPUT :


eclipse-workspace - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
> FactoryMethod [FactoryMethodPatt
JRE System Library [JavaSE-21]
> src
> factory
EmailFactory.java
EmailNotification.java
Main.java
Notification.java
NotificationFactory.java
SMSFactory.java
SMSNotification.java
module-info.java
Javadoc Console Progress History Sponsor the Eclipse IDE
<terminated> Main (1) [Java Application] C:\Users\Thanisha G\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (27-Jul-2026, 2:49:24 pm – 2:49:25 pm elapsed 0:00:00) [pid: 211
Sending SMS Notification